Inflation

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MEANING OF INFLATION

Inflation is commonly understood as a situation of substantial, and general increase in the level of prices of goods and services in an economy and a consequent fall in the value of money over a period of time.

When the general price level rises, value of money falls and as such each unit of currency buys fewer goods and services. Consequently, inflation reflects a reduction in the purchasing power per unit of money.

A chief measure of price inflation is the inflation rate which expresses percentage change in a general price index (normally the consumer price index) over time. Mathematically, rate of inflation can be expresses as:

$$\text{Rate of inflation} = \left(\frac{P_t - P_{t-1}}{P_{t-1}}\right) \times 100 \text{ percent}$$

Where P_t and P_{t-1} are price level at two time periods respectively. Price level is the average of prices.

SOME DEFINITIONS OF INFLATIONS

Monetarists' View: Monetarists assert that inflation has always been a monetary phenomenon. The quantity theory of money, simply stated, says that any change in the amount of money in a system will change the price level. This theory begins with the Fisher's equation of exchange: **M V = PT**

Where

- M represent total quantity of money
- V is velocity of circulation of money (i.e. average number of time each unit of money is spent for the purchase of goods and services during a given time period).
- P represents general price level
- T refers to the total volume of transactions (real value final goods and services)

Here MV represents supply of money
PT represents demand for money.

By manipulation, $P = \frac{M \quad V}{T}$

Assuming V and T as given, price level varies in directly in proportion to the quantity of money (M). Thus, if supply of money increases , there is inflation or rise in prices.

SOME DEFINITIONS OF INFLATIONS

Keyenes' View: Inflation occurs when price rises after the stage of full employment is reached in the economy, with no corresponding rise in employment and output.

Types of Inflation

There are five main types of inflation, categorized by their speed. They are creeping, walking, galloping and hyperinflation, stagflation . There are specific types of asset inflation and also wage inflation.

- **Creeping or mild inflation**

Creeping inflation Creeping or mild inflation is when prices rise three percent a year or less. This kind of mild inflation makes consumers expect that prices will keep going up. That boosts demand. Consumers buy now to beat higher future prices. That's how mild inflation drives economic expansion. The relatively small effect of creeping inflation, when viewed long-term, actually adds up to a pretty significant increase in the cost of living.

Walking or trotting inflation:

When the annual inflation rate is in between 3% to 10% , it is called walking or trotting inflation. Inflation at this rate is a warning signal for the government to control it before it turns into running inflation. Common goods and services are priced out of the reach of most people.

Galloping Inflation

When inflation rises to 10 percent or more, it wreaks absolute havoc on the economy. Money loses value so fast that business and employee income can't keep up with costs and prices. Foreign investors avoid the country, depriving it of needed capital. The economy becomes unstable, and government leaders lose credibility. Galloping inflation must be prevented at all costs.

Hyperinflation

Hyperinflation is when prices skyrocket more than 50 percent a month. It is very rare. In fact, most examples of hyperinflation have occurred only when governments printed money to pay for wars. Examples of hyperinflation include Germany in the 1920s, Zimbabwe in the 2000s, and Venezuela in the 2010s. The last time America experienced hyperinflation was during its civil war.

Types of Inflation continued.

Stagflation:

Stagflation is persistent high inflation combined with high unemployment and sluggish demand , and stagnant economy in a country. This seems contradictory. Economy is experiencing stagnant demand, still the prices are rising. It happened in the 1970s when the United States abandoned the gold standard.

AD = C+ I

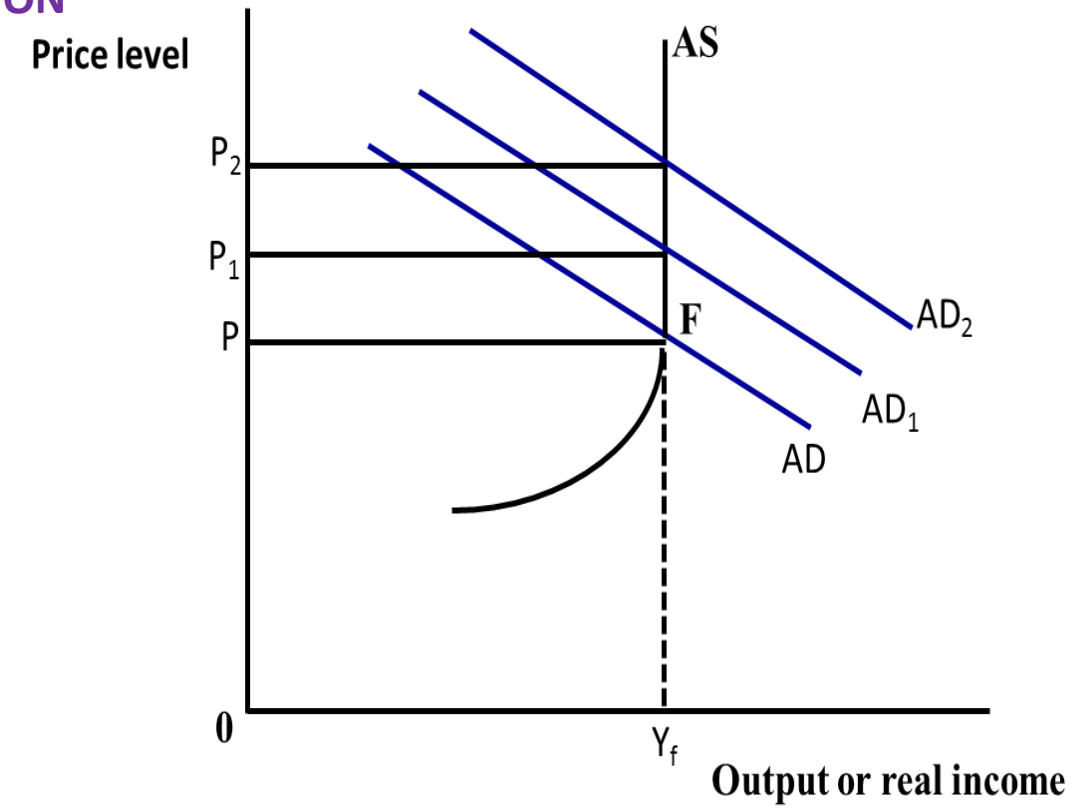
CAUSES OF INFLATION

Broadly speaking there are two school of thought regarding the possible causes of inflation.

- **Demand-pull inflation theory AD>AS**
- **Cost-push inflation theory**

DEMAND-PULL INFLATION

According to Demand-pull theory, price rises when aggregate_demand (AD) in an economy exceeds aggregate supply (AS) of goods and services at full employment level. The demand-pull theorists point out that inflation might be caused , in the first place, by an increase in in the quantity of money, when the economy is operating at full-employment. As the quantity of money increases, the rate of interest will fall and consequently investment will increase. The increase in investment expenditure will increase the income of various factors of production. As a result, aggregate consumption expenditure will increase leading to an increase in effective demand. With economy already operating at the level of full-employment, this will immediately raise prices, and inflationary forces may emerge. Thus, when general monetary demand rises faster than the general supply, it pulls up prices (both commodity as well as factor prices).



In figure, curves AD, AD1, and AD2 represents aggregate demand curves. The AS curve represent the aggregate supply function which slopes upward and becomes vertical straight line at point F indicating that the economy has reached full-employment.

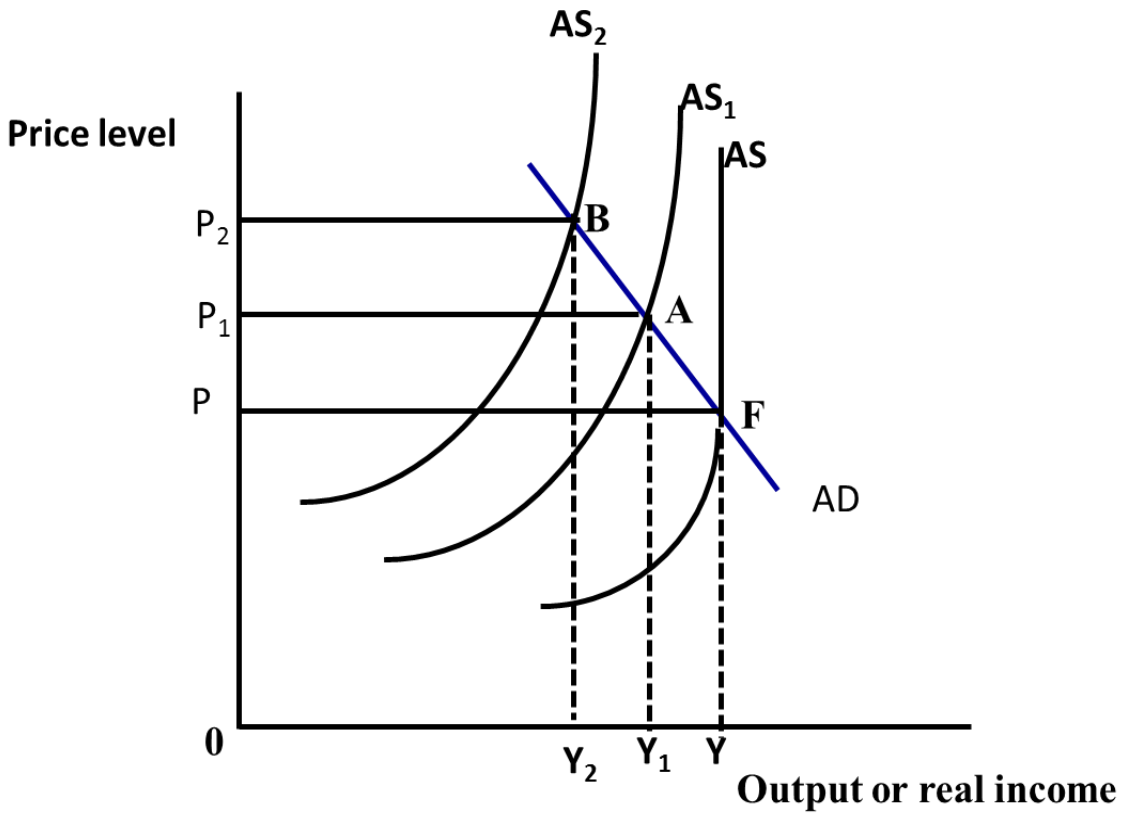
DEMAND-PULL INFLATION

Hence real output tend to be fixed or inelastic at point. Assuming that AD1 curve intersects with AS curve at point F, the real output or income is Y at full-employment and price level is P. when there is an increase in aggregate demand there would be an upward shift in aggregate demand curve such as from AD to AD1, or to AD2. The aggregate supply being inelastic, the price level then to rise from P to P1 and then to P2. There are many reasons which causes demand-pull inflation. Some of them are as follows are as follows:

- **Increase in autonomous investment in firms Increase in government spending or public expenditure**
- **A quick increase in consumption or MPC**
- **A sudden increase in exports which might lead to a huge under-valuation of your currency**
- **Excessive monetary growth – when they is too much money in the system chasing too few goods. The 'price' of a good will thus increase**

COST-PUSH INFLATION

Cost-push inflation occurs due to rising production costs. As production costs rises, the businessman raises the prices of their product in order to maintain their profit margins. Thus, cost push inflation is determined by supply side factors and can lead to lower economic growth and often causes a fall in living standards, though it often proves to be temporary. Higher costs shift a firm's supply curve upwards and lead to an increase in price. The phenomenon of cost-push inflation is graphically illustrated as under.



In figure, AD curve represents the aggregate demand curve. The AS, AS_1, AS_2 curves represents the aggregate supply function. AS curve becomes vertical or inelastic at point F which is the full-employment equilibrium point. The full-employment level of income is OY which can be maintained at price level P. Now if we begin with the price level P, F is the point of intersection AD and AS. If there is an increase in cost of production, the aggregate supply curve will shift upward from AS to AS_1, and consequently, the new equilibrium points will be A with real output OY_1 at higher price level P_1.

Now if we begin with the price level P, F is the point of intersection AD and AS. If there is an increase in cost of production, the aggregate supply curve will shift upward from AS to $AS_{1,}$ and consequently, the new equilibrium points will be A with real output OY_1 at higher price level P_1. A further rise in cost of production will cause shift in supply curve to AS_2 which will intersect with AD curve at point B and consequently price level will rise to P_2,real output will fall to $OY_{2,}$ Clearly the level of real output at A and B equilibrium points is less than full employment level. This means that with increase in cost of production, price level will rise cause fall in real output and rise in unemployment.

There are many reasons for cost-push inflation:

Rising prices of raw materials: Rising cost raw materials especially strategic raw materials like fuels etc. increase the , transportation cost, manufacturing cost and consequently cost of production and prices would rise would rise.

Rising labour costs: Wages are one of the main costs facing firms. Rising wages will push up prices as firms have to pay higher costs (higher wages may also cause rising demand) caused by increase in wage. Wages are on the rise due to increasing pressure from trade unions and labour unions to raise wages.

Higher indirect taxes: Higher indirect taxes imposed by the government on goods and services specially on essential commodities increase the cost of production and price.

Imported Inflation: Devaluation will increase the domestic price of imports. Therefore, after devaluation we often get an increase in inflation due to rising cost of imports.

Profit-push: when there are monopolies in the product market, monopoly firms would raise the price of the product in order to reap more profit. Consequently prices rise.

CONSEQUENCES OF INFLATION

High inflation rate may result in the following adverse effects on the economy:

Greater uncertainty: There may be greater uncertainty for both firms and households. Firms will postpone their investment due to uncertainty in the market. This will result in negative implications on the economic growth in the economy.

Redistributive effects: High rate of inflation will adversely affect people who have constant incomes, such as retired people, students, and dependents. Moreover, rise in prices of essential commodities (food & clothing) will affect the poor segment of the society as they spend a major part of their income on these good. This will lead to increased inequality in the economy.

Less saving: High rate of inflation will have an adverse effect on the savings in the economy. As people spend more to sustain their present standard of living, less is being saved. This will result in less loanable funds being available to firms for investment.

Business community: Entrepreneurs and businessman welcomes inflation as the stand to profit when price is rising. They find that the value of their inventories and stock of goods is rising in money terms.

Debtor and creditor: Debtors generally gain and creditors loose lose during inflation. Gain accrues to a borrowers when he repays loans when value of money has fallen due to inflation. Thus a borrowers pay less in real terms when he repays his loans during inflation. The creditor on the other hand is a loser since he receives less in real terms.

Damage to export competitiveness: High rate of inflation will hit hard the export industry in the economy. The cost of production will rise and the exports will become less competitive in the international market. Thus, inflation has an adverse effect on the balance of payments.

Social unrest: High rate of inflation leads to social unrest in the economy. There is increase dissatisfaction in among the workers as they demand higher wages to sustain their present living standard. Moreover, high rate of inflation leads to a general feeling of discomfort for the household as their purchasing power is consistently falling.

Interest rates: The Central Bank might use monetary tools to control high inflation rate by increasing interest rates. This will increase the cost of borrowing and will have a negative effect on both consumption and investment.

Deflation

Deflation is the opposite of inflation. It's when prices fall. Deflation causes demand deficiency , and may lead to economic crisis.